

Hybrid coupler types and phase splits

Type	Phase split	BW	Use
Wilkinson	0°	N/A	in-phase split
90° Hybrid	$0^\circ/90^\circ$	~20%	I/Q mixers, balanced amps
180° Rat-race	$0^\circ/180^\circ$	~20%	Balanced mixers, Σ/Δ
Lange Coupler	$0^\circ/90^\circ$	octave	Wideband balanced amps